WHEN IS THE ENESTRÖM–KAKEYA ANNULUS SHARP? EXACT EQUALITY CASES AND A DEFECT IDENTITY

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ABSTRACT. Let $p(z) = \sum_{k=0}^{n} a_k z^k$ with $a_k > 0$, and put $A = \max_k a_k/a_{k+1}$ and $a = \min_k a_k/a_{k+1}$. The Eneström–Kakeya theorem places all zeros of p in the annulus $a \le |z| \le A$. We determine exactly when this annulus is *sharp*, that is, when a zero actually lies on a boundary circle. In the standard rescaling $b_k = a_k A^k$ that makes the coefficient sequence nondecreasing, a zero on the unit circle exists if and only if b is constant on consecutive blocks of some common length $d \ge 2$ dividing $n+1$; equivalently, if and only if $q(u) = (1+u+\cdots+u^{d-1})G(u^d)$ for such a d. Three consequences are that every boundary zero is a root of unity of order dividing $n+1$, that for $n+1$ prime the outer circle is attained only by geometric polynomials, and that a simple arithmetic condition on the set of indices where the maximal ratio is *not* attained rules out boundary zeros. The proof is a rigidity argument for equality in the triangle inequality. It is complemented by an exact defect identity and by a worked perturbation showing that the extremal rate in the family $1+z+\cdots+z^{n-1}+\gamma z^n$ is $2\sin^2(\pi/(n+1))(\gamma-1)/(n+1) = \Theta((\gamma-1)/n^3)$. All statements are checked computationally, over 6182 coefficient sequences with no counterexample.

1. INTRODUCTION

1.1. The classical statement.

Let $p(z) = \sum_{k=0}^{n} a_k z^k$ with $a_k > 0$ for all k, and put

$$r_k := \frac{a_k}{a_{k+1}} \quad (0 \le k \le n-1), \qquad a := \min_{0\le k\le n-1} r_k, \qquad A := \max_{0\le k\le n-1} r_k. \tag{1}$$

The Eneström–Kakeya theorem states that every zero z of p satisfies

$$a \le |z| \le A. \tag{2}$$

The result is due to Eneström [E] and, independently and in the form above, to Kakeya [K]; see [M, §27] and [RS, Chapter 4] for the classical treatment and for refinements under additional hypotheses on the coefficients.

Two features of (2) are immediate. First, the annulus is optimal as stated: for $p(z) = 1 + z + \cdots + z^n$ all ratios equal 1 and all zeros lie on the unit circle. Second, no annulus determined by the extreme ratios alone can be smaller, since the ratios of a polynomial may be prescribed essentially freely.

The question studied here is different, and appears not to have been isolated in the literature: *for a given p, does a zero actually lie on the circle $|z| = A$, or on $|z| = a$?* That is, when is (2) attained rather than merely approached? The known refinements of (2) shrink the annulus under further hypotheses on the coefficients; they do not characterise the polynomials for which the original annulus is exact. Strictness has been studied for particular parametric families; the general characterisation below covers such families as special cases.

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1.2. **Main results.** The key step is the passage to the *extremal normalisation*. Set

$$b_k := a_k A^k \quad (0 \le k \le n), \qquad q(u) := \sum_{k=0}^{n} b_k u^k, \qquad p(z) = q(Az). \tag{3}$$

Since $b_k/b_{k+1} = r_k/A \le 1$, the sequence b is nondecreasing and positive, and the upper bound in (2) becomes the statement that the zeros of q lie in the closed unit disc. Write

$$\Delta_k := b_k - b_{k-1} \ge 0 \quad (1 \le k \le n), \qquad I := \{\, k \in \{1, \dots, n\} : \Delta_k > 0 \,\} \tag{4}$$

for the *jump set* of b, that is, the set of indices at which the maximal ratio A is not attained.

Theorem 1.1 (Rigidity)**.** *Let $0 < b_0 \le b_1 \le \cdots \le b_n$ and $q(u) = \sum_{k=0}^{n} b_k u^k$. The following are equivalent.*

(i) *q has a zero on the unit circle $\{|u| = 1\}$;*
(ii) *there is an integer $d \ge 2$ with $d \mid (n+1)$ such that $b_{jd} = b_{jd+1} = \cdots = b_{jd+d-1}$ for every $j = 0, 1, \dots, (n+1)/d - 1$;*
(iii) *there are $d \ge 2$ with $d \mid (n+1)$, an integer $m := (n+1)/d$, and numbers $0 < c_0 \le c_1 \le \cdots \le c_{m-1}$ such that*

$$q(u) = \bigl(1 + u + \cdots + u^{d-1}\bigr) \sum_{j=0}^{m-1} c_j u^{jd}. \tag{5}$$

Thus a boundary zero is never an accident of the coefficients: it forces a block-periodic structure, and it always arises from the cyclotomic factor $1 + u + \cdots + u^{d-1}$.

Theorem 1.1 yields, in §3.2, that every zero on $|u| = 1$ is a root of unity of order dividing $n+1$ (Corollary 3.3), that for $n+1$ prime the outer circle is attained only by geometric polynomials (Corollary 3.4), and that the existence of boundary zeros is decided by the arithmetic of the jump set (Corollary 3.5).

Section 4 gives the quantitative counterpart. There is an exact *defect identity* (Proposition 4.1), whose degenerate case is precisely the rigidity statement, and which controls the deficit of $|z|$ from A by the misalignment of the arguments u^k. A worked perturbation (Proposition 4.3) shows that in the family $1 + z + \cdots + z^{n-1} + \gamma z^n$ the extremal circle is approached at the rate $2\sin^2(\pi/(n+1))(\gamma - 1)/(n+1) = \Theta((\gamma - 1)/n^3)$: the Eneström–Kakeya bound is sharp, but its sharpness is fragile in high degree.

Section 5 reports an exhaustive verification of Theorem 1.1 over 6182 coefficient sequences, with no counterexample. Section 6 lists the open questions.

2. The sharp theorem and a short proof

For completeness, and because the proof of Theorem 1.1 builds on it, we recall a short proof of (2). The only ingredient is the following lemma, which is the case $A = 1$ of the theorem.

Lemma 2.1 (Monotone lemma)**.** *Let $0 < b_0 \le b_1 \le \cdots \le b_n$ and $q(u) = \sum_{k=0}^{n} b_k u^k$. Then every zero u of q satisfies $|u| \le 1$.*

Proof. Expanding $(1 - u)q(u)$ and collecting the coefficients of u^k gives, for every u,

$$(1 - u)q(u) = b_0 + \sum_{k=1}^{n} \Delta_k u^k - b_n u^{n+1}, \qquad \Delta_k = b_k - b_{k-1} \ge 0.$$

Suppose that $q(u) = 0$ and $R := |u| > 1$. Then $b_n u^{n+1} = b_0 + \sum_{k=1}^{n} \Delta_k u^k$, hence

$$b_n R^{n+1} \le b_0 + \sum_{k=1}^{n} \Delta_k R^k \le b_0 + R^n \sum_{k=1}^{n} \Delta_k = b_0 + R^n(b_n - b_0) = b_n R^n - b_0\big(R^n - 1\big) < b_n R^n,$$

because $b_0 > 0$ and $R > 1$. Dividing by $b_n R^n > 0$ gives $R < 1$, a contradiction. □

Theorem 2.2 (Eneström–Kakeya)**.** *Every zero z of p satisfies $a \le |z| \le A$.*

Proof. Upper bound. With $b_k = a_k A^k$ as in (3) we have $p(z) = q(Az)$, where q has nondecreasing positive coefficients. By Lemma 2.1 every zero u of q satisfies $|u| \le 1$, so every zero $z = A^{-1}u$ of p satisfies $|z| \le A$.

Lower bound. Put $c_k := a_k a^k$, so that $c_k/c_{k+1} = r_k/a \ge 1$ and hence $c_0 \ge c_1 \ge \cdots \ge c_n > 0$. Then $p(z) = f(z/a)$ with $f(w) = \sum_{k=0}^{n} c_k w^k$. Let $f^*(w) := w^n f(1/w) = \sum_{j=0}^{n} c_{n-j} w^j$, whose coefficients $c_n \le c_{n-1} \le \cdots \le c_0$ are positive and nondecreasing; by Lemma 2.1 every zero w' of f^* satisfies $|w'| \le 1$. A non-zero w is a zero of f if and only if $1/w$ is a zero of f^*, so $|w| \ge 1$ for every zero w of f, and $|z| = a|w| \ge a$. □

Remark 2.3. The annulus (2) cannot be improved in general, as $p(z) = 1 + z + \cdots + z^n$ shows, where $a = A = 1$ and all zeros lie on $|z| = 1$. The content of §3 is that, for a given p with $a < A$, attainment on a boundary circle is an extremely restrictive condition.

3. Exact equality cases

We keep the notation (3)–(4): the sequence b is positive and nondecreasing, $q(u) = \sum b_k u^k$, and I is the jump set.

Lemma 3.1 (Decomposition identity)**.** *For every $u \in \mathbb{C}$,*

$$(1-u)q(u) = b_0 + \sum_{k=1}^{n} \Delta_k u^k - b_n u^{n+1}. \tag{6}$$

In particular, if $q(u) = 0$ then

$$b_0 + \sum_{k \in I} \Delta_k u^k = b_n u^{n+1}. \tag{7}$$

Proof. Expand $(1-u)\sum_{k=0}^{n} b_k u^k = \sum_{k=0}^{n} b_k u^k - \sum_{k=0}^{n} b_k u^{k+1}$ and collect the coefficient of u^k for $1 \le k \le n$, which is $b_k - b_{k-1} = \Delta_k$; the constant term is b_0 and the term of degree $n+1$ is $-b_n$. Terms with $\Delta_k = 0$ may be dropped, which gives (7). □

3.1. Proof of the rigidity theorem.

Proof of Theorem 1.1. (i)⇒(ii). Let $q(u) = 0$ with $|u| = 1$. Since $q(1) = \sum_k b_k > 0$ we have $u \ne 1$. By (7),

$$b_0 + \sum_{k \in I} \Delta_k u^k = b_n u^{n+1}. \tag{8}$$

Taking absolute values and using $|u| = 1$ together with $\sum_{k \in I} \Delta_k = b_n - b_0$,

$$b_n = \Big|b_0 + \sum_{k \in I} \Delta_k u^k\Big| \le b_0 + \sum_{k \in I} \Delta_k = b_n.$$

Hence equality holds in the triangle inequality. All summands b_0 and $\Delta_k u^k$ ($k \in I$) are nonzero, so they share a common argument; the summand $b_0 > 0$ has argument 0, hence

$$u^k = 1 \qquad \text{for every } k \in I. \tag{9}$$

Substituting (9) into (8) gives $b_0 + \sum_{k\in I} \Delta_k = b_n u^{n+1}$, that is $b_n = b_n u^{n+1}$, so

$$u^{n+1} = 1. \tag{10}$$

Set $d := \operatorname{ord}(u) \geq 2$. By (9) and (10) we have $d \mid k$ for all $k \in I$, and $d \mid (n+1)$. Since I is exactly the set of indices at which b strictly increases, b is constant between consecutive elements of I; as all elements of I are multiples of d, the sequence b is constant on every block $\{jd, \dots, jd + d - 1\}$. Because $d \mid (n+1)$, all blocks are complete, which is (ii).

(ii)⇒(iii). If b is constant on blocks of length d and $d \mid (n+1)$, put $c_j := b_{jd}$ for $0 \leq j \leq m-1$, $m = (n+1)/d$. Then $0 < c_0 \leq \cdots \leq c_{m-1}$ and

$$q(u) = \sum_{j=0}^{m-1} c_j \sum_{r=0}^{d-1} u^{jd+r} = \Big(\sum_{r=0}^{d-1} u^r\Big) \sum_{j=0}^{m-1} c_j u^{jd},$$

which is (5).

(iii)⇒(i). Let $u \neq 1$ be a d-th root of unity. Then $|u| = 1$, $1 + u + \cdots + u^{d-1} = 0$ and $u^d = 1$, so $q(u) = 0 \cdot \sum_j c_j = 0$. □

Remark 3.2. Only the implication (i)⇒(ii) uses monotonicity, and it uses it exactly twice: to make the coefficients Δ_k non-negative in (7), and to convert the alignment (9) into the block structure of b. For arbitrary positive coefficients the conclusion fails, because then the triangle inequality in the proof is generically strict.

3.2. **Corollaries.**

Corollary 3.3 (Boundary zeros are roots of unity)**.** *If $q(u) = 0$ and $|u| = 1$, then $u^{n+1} = 1$. Moreover q vanishes at every non-trivial d-th root of unity, for each $d \geq 2$ dividing $n+1$ that occurs in Theorem 1.1(ii).*

Proof. The first assertion is (10); the second is the implication (ii)⇒(i) of Theorem 1.1. □

Corollary 3.4 (Prime $n+1$)**.** *Suppose that $n+1$ is prime. Then the circle $|z| = A$ contains a zero of p if and only if all ratios r_k coincide, that is $a = A$ and $p(z) = c\sum_{k=0}^{n}(z/A)^k$ for some $c > 0$. In that case (2) degenerates to a single circle.*

Proof. If a zero with $|z| = A$ exists, Theorem 1.1 provides $d \geq 2$ with $d \mid (n+1)$; primality forces $d = n+1$, so b is constant on the single block $\{0, \dots, n\}$, i.e. $b_0 = \cdots = b_n$. Then $b_k = b_{k+1}$ gives $r_k = A$ for all k, hence $a = A$ and $a_k = a_0 A^{-k}$. Conversely, if $a_k = a_0 A^{-k}$ then $p(z) = a_0 \sum_k (z/A)^k$, whose zeros satisfy $(z/A)^{n+1} = 1$ with $z \neq A$, hence $|z| = A$. □

Corollary 3.5 (Arithmetic criterion)**.** *Let I be the jump set (4) and let g be the greatest common divisor of the elements of I (with $g := 0$ if $I = \emptyset$).*

(a) *If $I = \emptyset$, equivalently $b_0 = \cdots = b_n$ and $a = A$, then q vanishes at all $(n+1)$-th roots of unity other than 1.*

(b) *If $I \neq \emptyset$, then q has a zero on the unit circle if and only if $\gcd(g, n+1) \geq 2$.*

In particular, if the maximal ratio A is attained at a single index k_0 and $\gcd(k_0, n+1) = 1$, then no zero of p lies on $|z| = A$.

Proof. Part (a) is the case $d = n+1$ of Theorem 1.1, in which the single block covers all coefficients. For (b), Theorem 1.1 says that a boundary zero exists if and only if b is constant on blocks of some length $d \geq 2$ with $d \mid (n+1)$; this holds if and only if d divides every element of I, that is $d \mid g$. Such a $d \geq 2$ exists if and only if $\gcd(g, n+1) \geq 2$. The last assertion is the case $I = \{k_0\}$. □

Example 3.6.

(a) $q(u) = 1 + u + 2u^2 + 2u^3$, so $n+1 = 4$, $I = \{2\}$ and $\gcd(2,4) = 2$. Boundary zeros exist: indeed $q(u) = (1+u)(1+2u^2)$, and $u = -1$ has modulus 1. The remaining zeros $\pm \mathrm{i}/\sqrt{2}$ have modulus $1/\sqrt{2} < 1$, so the outer circle is attained and the inner one is not.

(b) $q(u) = 1 + u + u^2 + \gamma u^3$ with $\gamma > 1$, so $n+1 = 4$ and $I = \{3\}$. Since $\gcd(3,4) = 1$, there is no zero of modulus 1: all zeros lie strictly inside the unit disc. Note that $b = (1,1,1,\gamma)$ is *almost* block-periodic with $d = 2$, the only failure being in the last block, and that this already destroys attainment.

(c) $q(u) = 1 + u + u^2 + u^3 + 2u^4$, so $n+1 = 5$ is prime. There is no zero on the unit circle, in agreement with Corollary 3.4.

(d) Take $b = (1,1,2,3,3,3)$, that is $q(u) = 1+u+2u^2+3u^3+3u^4+3u^5$, with $n+1 = 6$. Here $I = \{2,3\}$ and $\gcd(\gcd(2,3),6) = 1$, so there is no zero of modulus 1; numerically the maximal modulus is $0.878335\ldots$. Note that b is a small perturbation of $(1,1,1,3,3,3)$, which *is* block-periodic with $d = 3$ and does attain the bound. A single misplaced coefficient therefore destroys attainment, in agreement with the rigidity of Theorem 1.1.

Remark 3.7 (The inner circle). Attainment of the lower bound is governed by the reversed sequence. With $c_k = a_k a^k$ and $f(w) = \sum_k c_k w^k$ as in the proof of Theorem 2.2, a zero z of p satisfies $|z| = a$ if and only if the zero $w = z/a$ of f satisfies $|w| = 1$, that is, if and only if $f^*(w) = \sum_j c_{n-j} w^j$ satisfies the hypothesis of Theorem 1.1 with c_{n-j} in place of b_j. Reversing a block structure of length d with $d \mid (n+1)$ produces a block structure of the same length, so the two conditions have the same shape, but they are imposed on the two different rescalings $a_k A^k$ and $a_k a^k$; a sequence may well attain one bound and not the other, as in Example 3.6(a).

4. The quantitative side

Theorem 1.1 describes the exact equality case. The next two results quantify what happens nearby: an identity that measures the deficit, and a computation that shows how fast the extremal circle is approached under a natural perturbation.

4.1. **A defect identity.**

Proposition 4.1 (Defect identity)**.** *Let $0 < b_0 \le \cdots \le b_n$ and let u be a zero of q. Then*

$$b_n\bigl(1 - \operatorname{Re} u^{n+1}\bigr) = \sum_{k=1}^{n} \Delta_k \bigl(1 - \operatorname{Re} u^k\bigr). \tag{11}$$

Consequently, with $R = |u|$ and $u = Re^{\mathrm{i}\vartheta}$,

$$\sum_{k\in I} \Delta_k \bigl(1 - R^k \cos(k\vartheta)\bigr) \;\ge\; b_n\bigl(1 - R^{n+1}\bigr). \tag{12}$$

Proof. Taking real parts in (7) and using $\sum_{k\in I} \Delta_k = b_n - b_0$ gives

$$b_n \operatorname{Re} u^{n+1} = b_0 + \sum_{k\in I} \Delta_k \operatorname{Re} u^k = b_n - \sum_{k\in I} \Delta_k + \sum_{k\in I} \Delta_k \operatorname{Re} u^k,$$

which is (11). For (12), use $\operatorname{Re} u^{n+1} \le R^{n+1}$ in the left-hand side of (11) and $\operatorname{Re} u^k = R^k \cos(k\vartheta)$ in the right-hand side. □

Remark 4.2. At $R = 1$ the identity (11) reads $b_n\bigl(1 - \cos((n+1)\vartheta)\bigr) = \sum_{k\in I} \Delta_k \bigl(1 - \cos(k\vartheta)\bigr)$, with both sides non-negative. This is a "linearised" form of the rigidity argument, but it is

strictly weaker: the real part alone does not force the individual terms to vanish, which is why the alignment (9) was deduced from the modulus rather than from (11).

4.2. **How fast is the extremal circle approached?** Fix $n \geq 1$ and consider the family

$$p_\gamma(z) = 1 + z + \cdots + z^{n-1} + \gamma z^n, \qquad \gamma > 1. \tag{13}$$

Its ratios are $r_0 = \cdots = r_{n-2} = 1$ and $r_{n-1} = 1/\gamma$, so $A = 1$, $a = 1/\gamma$ and the jump set is $I = \{n\}$. By Corollary 3.5(b) there is no zero of modulus 1, since $\gcd(n, n+1) = 1$. It is therefore natural to ask how far the zeros are from the unit circle.

Proposition 4.3 (Sharpness rate)**.** *Let $\omega = e^{2\pi \mathrm{i} k/(n+1)}$ with $1 \leq k \leq n$, and let $z_k(\gamma)$ denote the zero of p_γ near ω. Then z_k is analytic near $\gamma = 1$ and*

$$z_k(\gamma) = \omega + \frac{1-\omega}{n+1}(\gamma - 1) + O\big((\gamma-1)^2\big), \tag{14}$$

$$|z_k(\gamma)| = 1 - \frac{2\sin^2\big(\pi k/(n+1)\big)}{n+1}(\gamma - 1) + O\big((\gamma-1)^2\big). \tag{15}$$

In particular

$$\max_{1\leq k\leq n} |z_k(\gamma)| = 1 - \frac{2\sin^2\big(\pi/(n+1)\big)}{n+1}(\gamma - 1) + O\big((\gamma-1)^2\big) \tag{16}$$

$$= 1 - \frac{2\pi^2}{(n+1)^3}(\gamma - 1) + O\big((\gamma-1)^2\big) + O\left(\frac{\gamma-1}{(n+1)^5}\right). \tag{17}$$

Proof. Write $p_\gamma(z) = \dfrac{1-z^{n+1}}{1-z} + (\gamma-1)z^n$ and set $\varepsilon = \gamma - 1$. At $\varepsilon = 0$ the zeros are the $(n+1)$-th roots of unity $\omega \neq 1$, all simple, so by the implicit function theorem each of them continues analytically to a zero $z_k(\varepsilon)$ of p_γ. Differentiating $p_\gamma(z_k(\varepsilon)) = 0$ at $\varepsilon = 0$ gives $z_k'(0) = -\omega^n / p_1'(\omega)$. Now $p_1'(z) = \sum_{k=1}^{n} k z^{k-1} = \dfrac{1 - (n+1)z^n + n z^{n+1}}{(1-z)^2}$, so at $z = \omega$, using $\omega^{n+1} = 1$ and $\omega \neq 1$,

$$p_1'(\omega) = \frac{1 - (n+1)\omega^{-1} + n}{(1-\omega)^2} = \frac{(n+1)\big(1-\omega^{-1}\big)}{(1-\omega)^2} = -\frac{(n+1)\omega^{-1}}{1-\omega},$$

and therefore

$$z_k'(0) = -\frac{\omega^n}{p_1'(\omega)} = \omega^n \cdot \frac{1-\omega}{(n+1)\omega^{-1}} = \frac{\omega^{n+1}(1-\omega)}{n+1} = \frac{1-\omega}{n+1},$$

which is (14). Then

$$|z_k(\varepsilon)|^2 = 1 + \frac{2\varepsilon}{n+1}\operatorname{Re}\big(\overline{\omega}(1-\omega)\big) + O(\varepsilon^2) = 1 - \frac{4\varepsilon}{n+1}\sin^2\Big(\frac{\pi k}{n+1}\Big) + O(\varepsilon^2),$$

since $\operatorname{Re}\big(\overline{\omega}(1-\omega)\big) = \cos\vartheta - 1 = -2\sin^2(\vartheta/2)$ with $\vartheta = 2\pi k/(n+1)$. Taking square roots gives (15). The function $k \mapsto \sin^2(\pi k/(n+1))$ is maximal over $1 \leq k \leq n$ at $k = 1$, which gives the first equality in (16); the second follows from $\sin t = t + O(t^3)$. □

Remark 4.4. Proposition 4.3 makes the fragility of (2) quantitative: a relative perturbation of size ε moves the extremal zeros off the circle by only $\Theta(\varepsilon/n^3)$, so forcing a zero to lie a distance δ inside the disc requires $\gamma - 1 \asymp n^3\delta$. The exponent 3 reflects the double degeneracy at $u = 1$, where both the block structure and the $(n+1)$-th root of unity condition collapse.

TABLE 1. Proposition 4.3 against direct computation of the maximal root modulus of (13).

n	γ	$\max_k \lvert z_k(\gamma)\rvert$	first-order prediction
4	1.001	0.999861815004	0.999861803399
4	1.1	0.986330205396	0.986180339887
9	1.001	0.999980891439	0.999980901699
9	1.1	0.997994875902	0.998090169944
19	1.001	0.999997550857	0.999997552826
19	1.1	0.999736088306	0.999755282581

5. Computational verification

All statements of §3 and §4 were checked with the scripts in the companion repository (`verify/`); the entry points are `ek_core.py`, `ek_more.py` and `ek_dual.py`.

Method. For an integer sequence b, the question whether q has a zero on $|u| = 1$ is decided as follows. Since q has real coefficients, a zero u with $|u| = 1$ satisfies $q(u) = u^n\overline{q(u)} = q^*(u)$, where $q^*(u) = u^n q(1/u)$ is the reversed polynomial. Hence a boundary zero forces the gcd $\gcd(q, q^*)$ to be non-constant, and this is decided *exactly* in rational arithmetic: when $\gcd(q, q^*) = 1$ there is no boundary zero and no floating-point computation enters the decision. When the gcd is non-constant, its zeros are computed numerically and their moduli inspected; in the search below this second step always produced a root of unity, in agreement with Corollary 3.3.

Exhaustive search. We enumerated all nondecreasing sequences $1 \le b_0 \le \cdots \le b_n \le 5$ with $2 \le n + 1 \le 12$, that is 6182 sequences, and compared the exact decision with the block-structure criterion of Theorem 1.1(ii). There were 591 sequences with a boundary zero, all of them block-structured, and no counterexample in either direction. In the same sweep, all 1111 zeros found on the unit circle satisfied $u^{n+1} = 1$, confirming Corollary 3.3.

The two-sided question. Among the 6116 sequences of the sweep with $a < A$, that is with a non-degenerate annulus, none attained both boundary circles. This is consistent with, but does not prove, the expectation discussed in Problem 6.2.

Defect identity. The identity (11) was verified numerically on 2000 random polynomials with positive rational coefficients of degree at most 8; the maximal relative error was $1.2 \cdot 10^{-14}$, at the level of round-off. Table 1 confirms Proposition 4.3.

6. Open questions

Problem 6.1 (Quantitative rigidity)**.** Find an explicit functional η of the coefficient sequence, vanishing exactly on the block-structured sequences of Theorem 1.1(ii), such that every zero u of q satisfies $|u| \le 1 - \eta(b)$ for a non-degenerate annulus. Proposition 4.3 shows that the true deficit can be as small as $\Theta(\varepsilon/n^3)$ for a perturbation of relative size ε, which suggests that η must be cubic in the distance to block structure and that a linear stability estimate cannot hold.

Problem 6.2 (Simultaneous attainment)**.** Suppose that $a < A$ and that both circles $|z| = a$ and $|z| = A$ contain a zero of p. Must then $a = A$? The search of §5 found no counterexample. The difficulty is that the two conditions of Theorem 1.1 are imposed on the two different rescalings $a_k A^k$ and $a_k a^k$ of the same coefficient sequence.

Problem 6.3 (Other settings)**.** The rigidity argument uses only the ordered field structure and the triangle inequality, so analogues should hold for the quaternionic, octonionic and matrix versions of the Eneström–Kakeya theorem that have been studied in the literature. A discrete analogue for the polynomial-method bounds on Kakeya sets in finite fields also seems plausible. We did not pursue either direction.

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